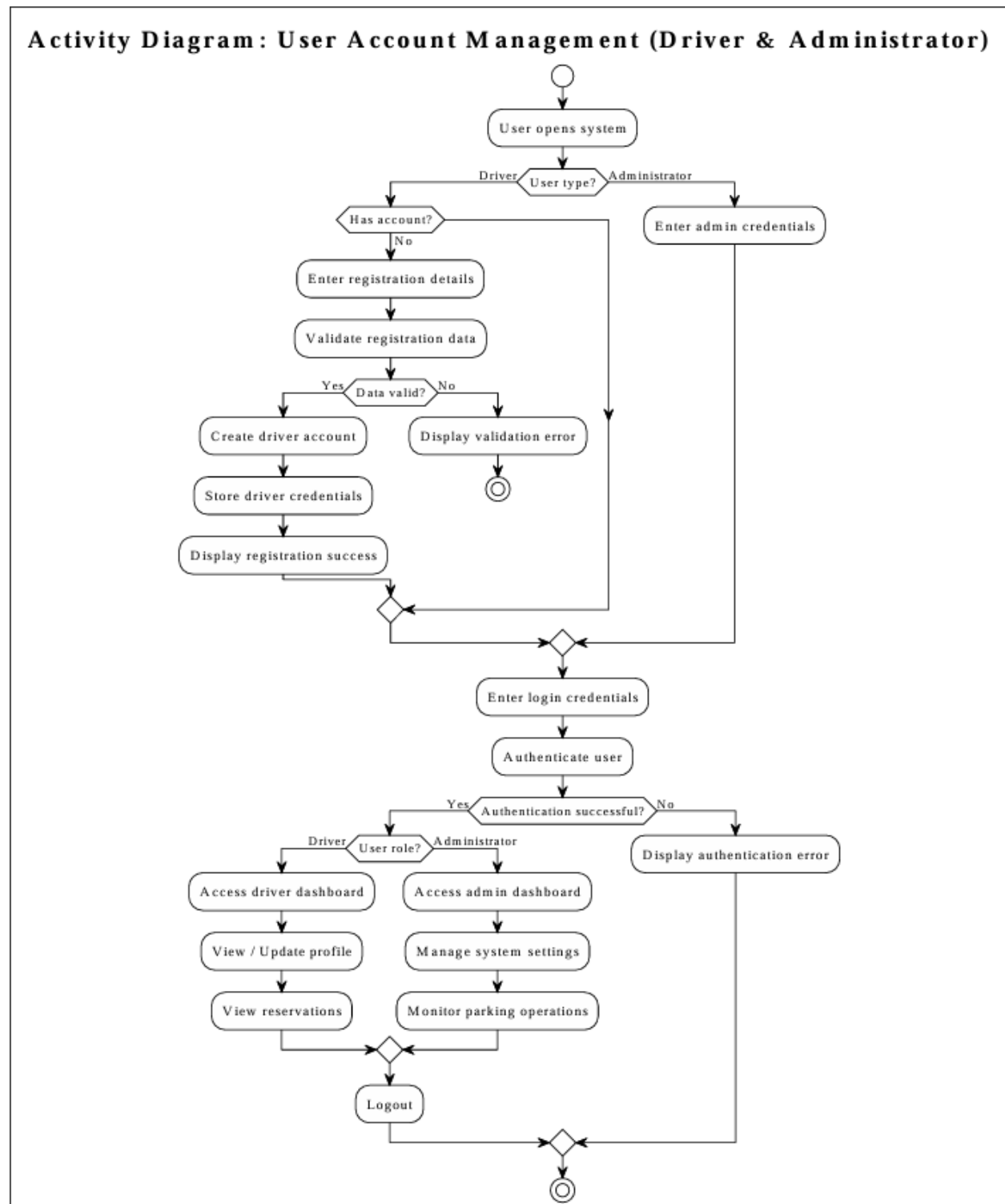


Diagram Explanations

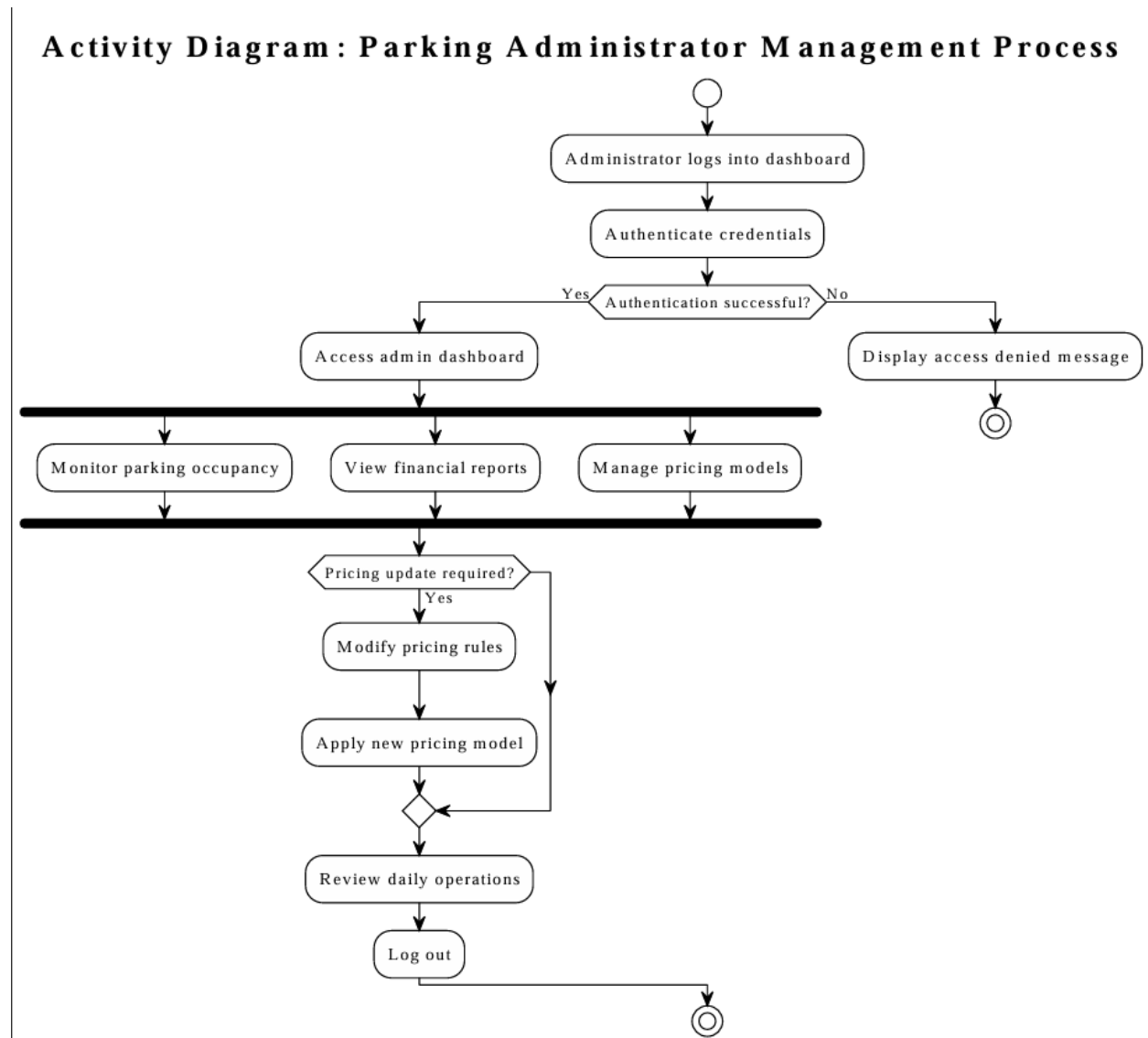
1. Activity Diagram: User Account Management (Driver & Administrator)



This diagram illustrates the authentication and onboarding workflow for the two primary system actors: Drivers and Administrators.

- **Process Flow:** The process begins when a user opens the system. The system first checks the user type.
- **Driver Path:** If the user is a driver, they can choose to register if they do not have an account. The system validates their registration data; if valid, a new account is created. Once registered, the driver logs in to access features like viewing their profile and checking reservations.
- **Administrator Path:** Administrators skip registration and proceed directly to login using pre-assigned credentials to access the admin dashboard, where they can manage settings and monitor operations.
- **Outcome:** The process ends when the user logs out or if authentication fails.

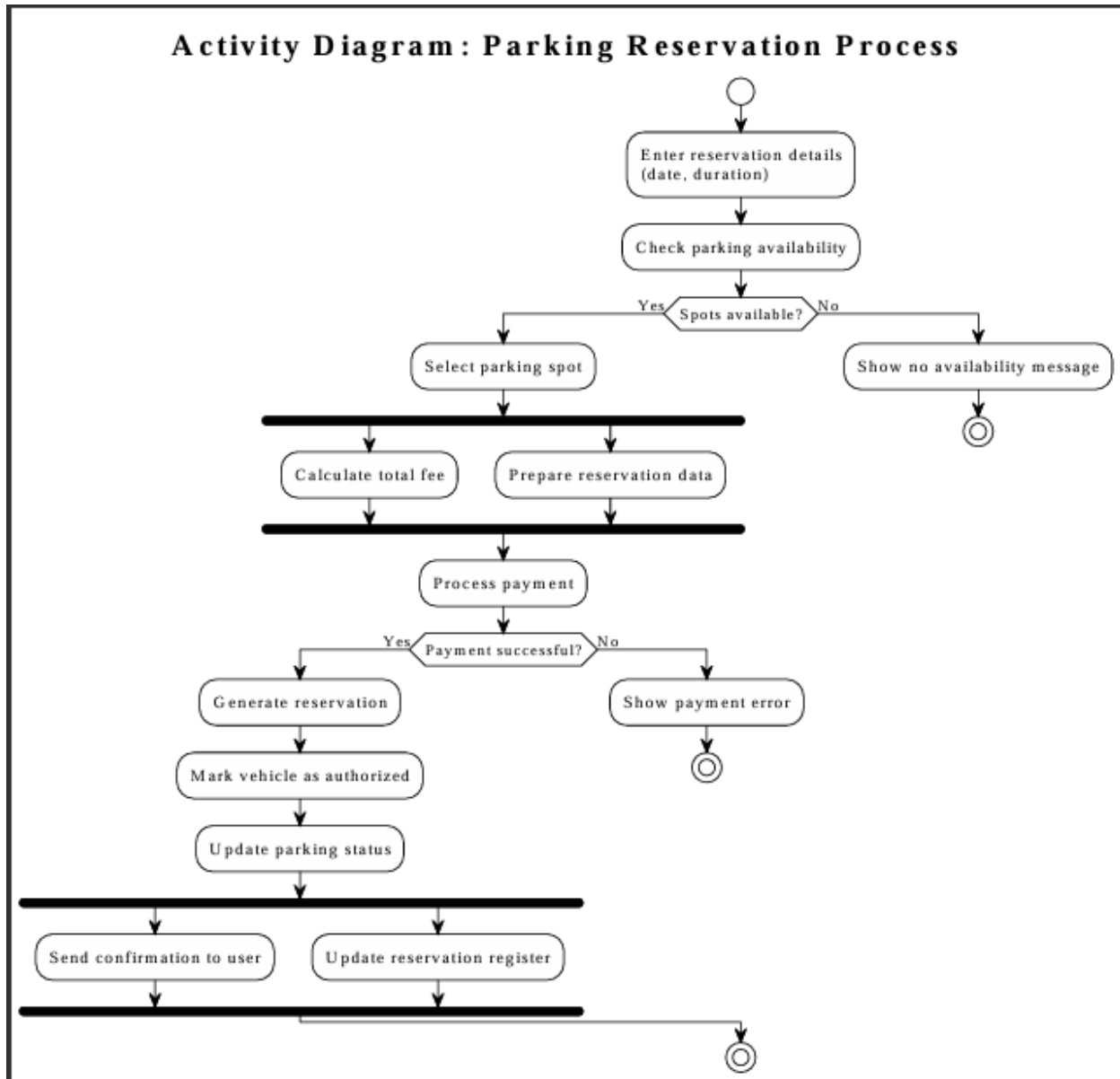
2. Activity Diagram: Parking Administrator Management Process



This diagram details the operational workflows available to the Parking Administrator after logging into the dashboard.

- **Process Flow:** Upon successful authentication, the administrator enters a parallel processing node (fork), allowing them to perform three simultaneous tasks: monitoring parking occupancy, viewing financial reports, and managing pricing models.
- **Key Decisions:** A specific decision point exists for pricing: if an update is required, the administrator modifies the rules and applies the new model.
- **Outcome:** The workflow concludes after the administrator reviews daily operations and logs out of the system.

3. Activity Diagram: Parking Reservation Process

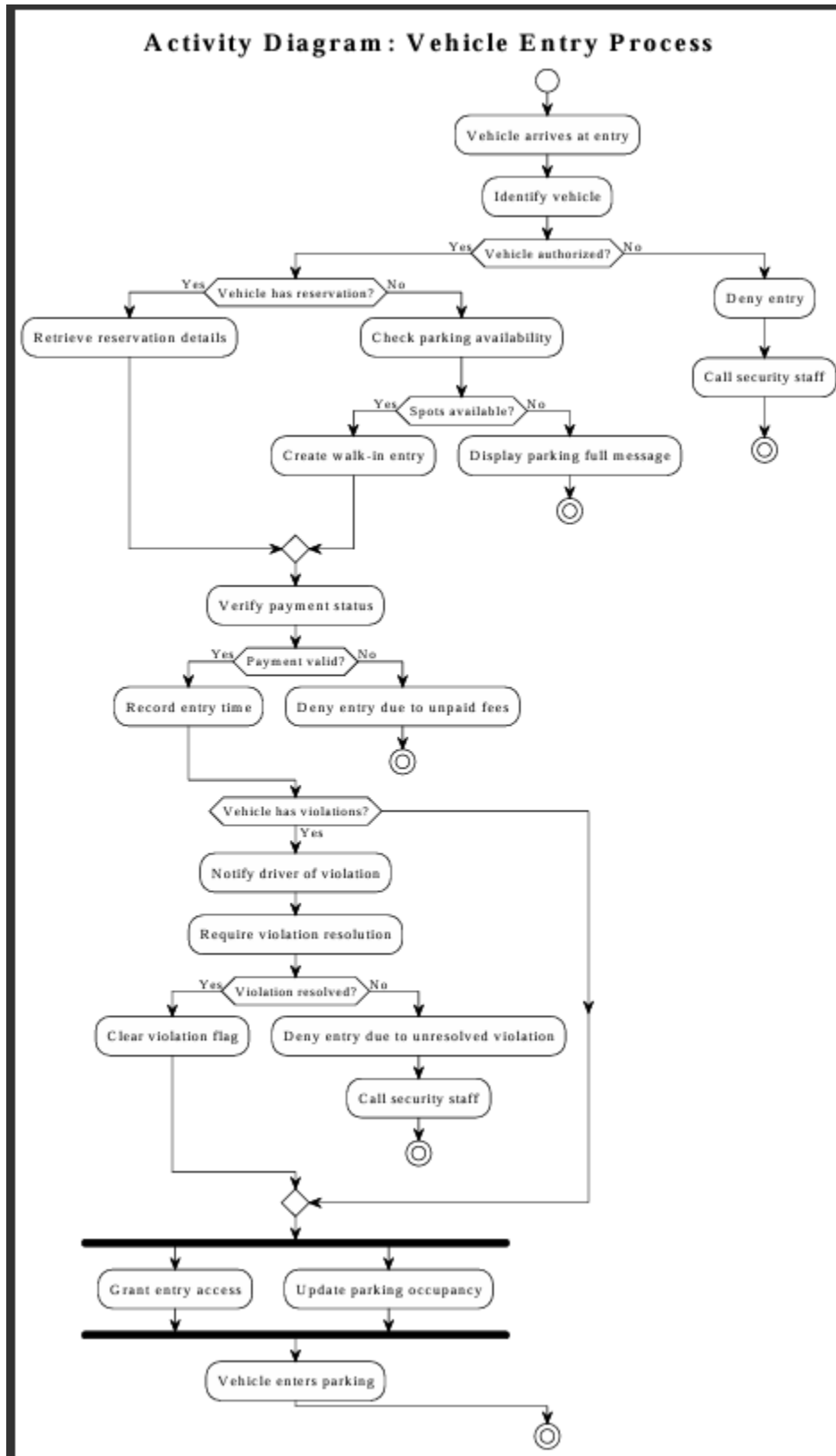


This diagram maps out the steps a driver takes to book a parking spot in advance via the mobile app.

- **Process Flow:** The driver enters reservation details (date, duration), and the system checks database availability. If no spots are free, the process ends with a message.
- **Booking Logic:** If a spot is available, the user selects it. The system then splits into parallel tasks to calculate fees and prepare reservation data.

- **Payment & Confirmation:** The driver processes the payment. If successful, the system generates the reservation, marks the vehicle as authorized, and simultaneously sends a confirmation to the user while updating the internal reservation register.

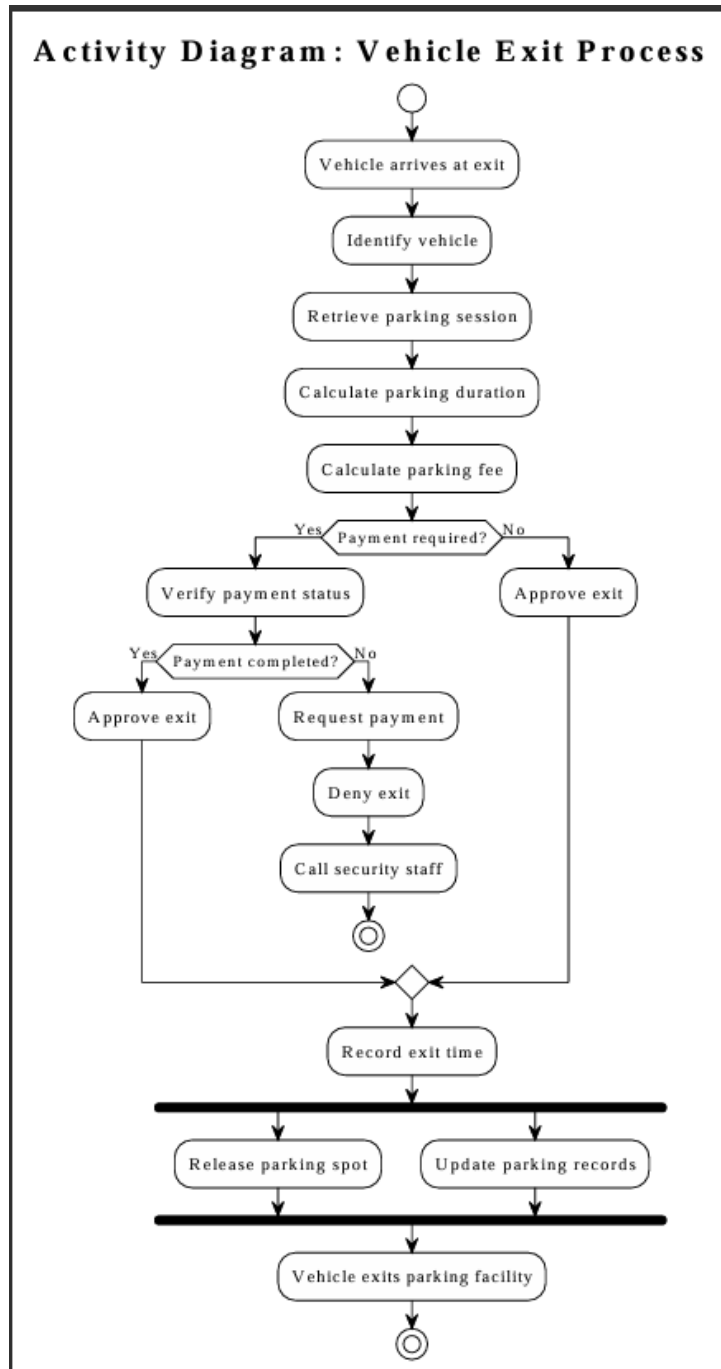
4. Activity Diagram: Vehicle Entry Process



This diagram captures the complex logic executed by the system when a vehicle arrives at the entry gate, handling both reserved and walk-in users.

- **Identification & Authorization:** The system identifies the vehicle (via ANPR). It first checks if the vehicle is authorized. If not, entry is denied, and security is notified.
- **Reservation vs. Walk-in:** If the vehicle is authorized, the system checks for an existing reservation. If none exists, it checks for walk-in availability. If the lot is full, the vehicle is turned away.
- **Validation:** The system performs checks for valid payment status and unresolved violations. If a violation is found, the driver must resolve it immediately, or entry is denied.
- **Outcome:** If all checks pass, the system performs a parallel action to grant access (open gate) and update the occupancy count.

5. Activity Diagram: Vehicle Exit Process

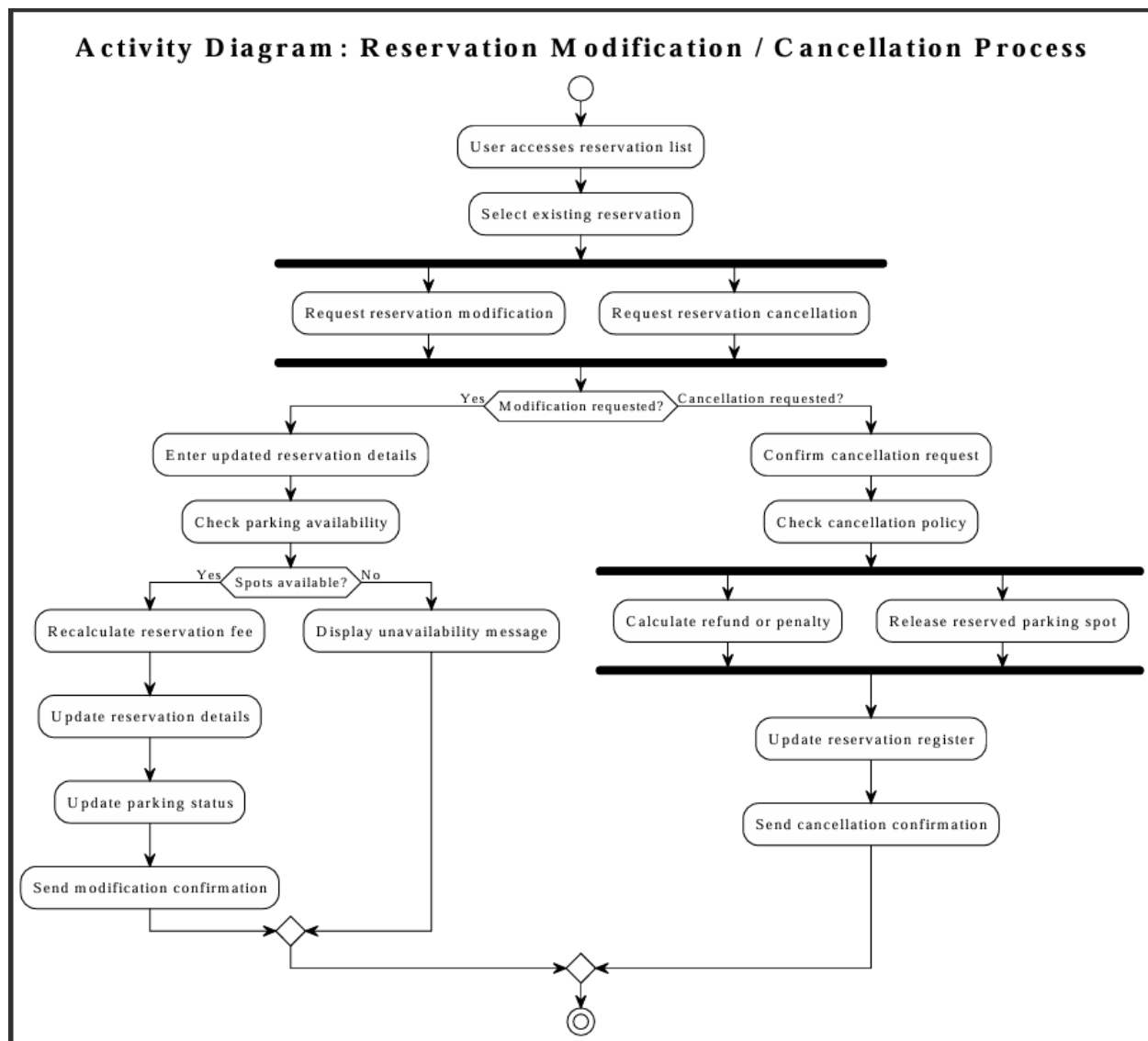


This diagram defines the logic for a vehicle leaving the facility, ensuring all fees are settled before the gate opens.

- **Process Flow:** When a vehicle arrives at the exit, the system identifies it and retrieves the active parking session to calculate the duration and total fee.

- **Payment Check:** The system determines if payment is required. If the user has already paid (or has a pre-paid reservation), exit is approved. If payment is pending, the system requests it; failure to pay results in a denied exit and a security alert.
- **Outcome:** Upon approval, the system records the exit time and performs parallel updates to release the parking spot in the database and archive the parking record.

6. Activity Diagram: Reservation Modification & Cancellation Process

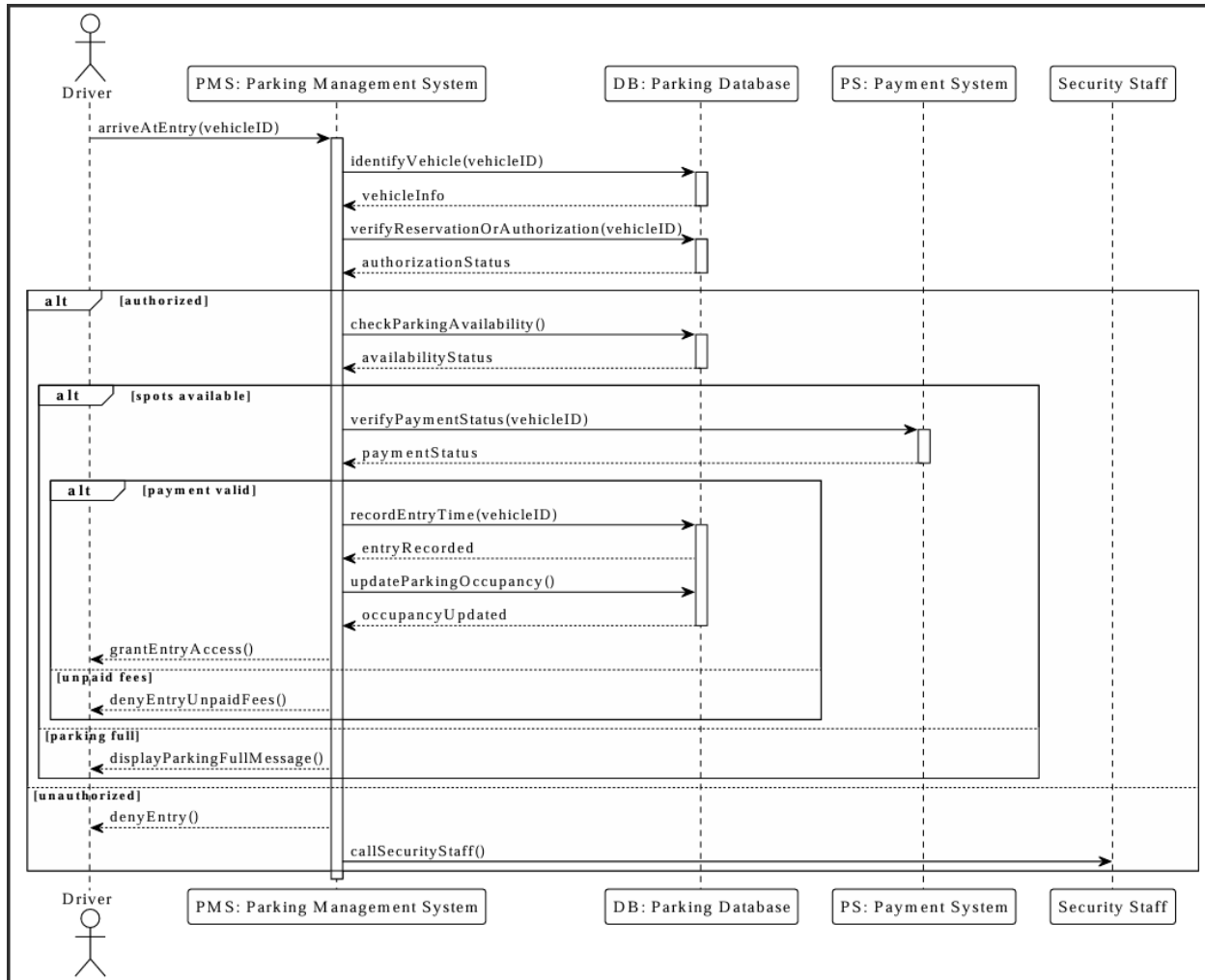


This diagram shows how users manage existing reservations, offering two distinct paths: modification or cancellation.

- **Modification Path:** If a user chooses to modify, the system accepts new details, checks availability for the new slot, recalculates fees, and updates the booking if successful.

- **Cancellation Path:** If cancellation is chosen, the system checks the policy. It then executes parallel actions to calculate any necessary refunds (or penalties) and release the reserved spot back to the pool of available parking.

7. Sequence Diagram: Vehicle Entry

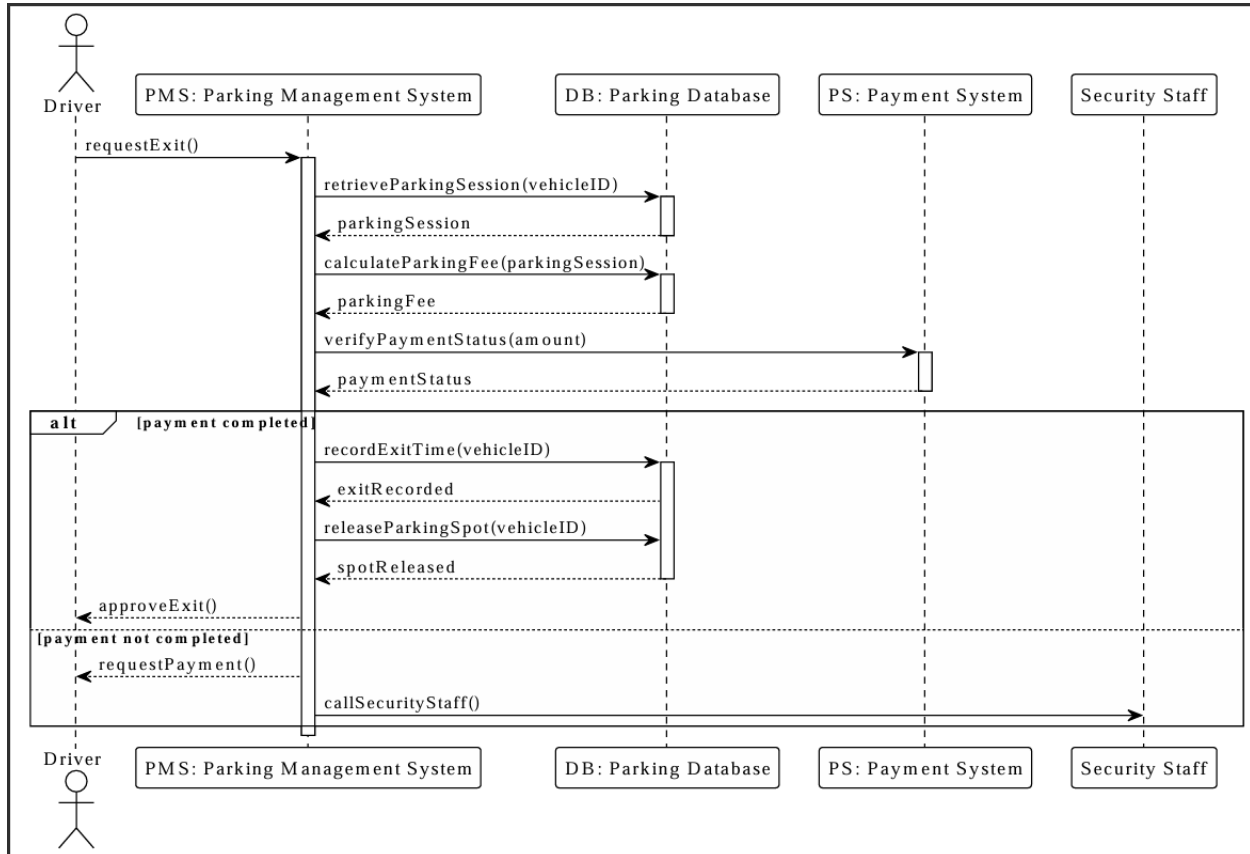


This diagram visualizes the real-time interactions between the Driver, Parking Management System (PMS), Database (DB), Payment System (PS), and Security Staff during entry.

- **Interaction:** The Driver triggers the flow by arriving. The PMS queries the DB to identify the vehicle and verify authorization.
- **Logic:** The PMS checks availability and then queries the PS to verify payment status.
- **Alt Frames:** The diagram uses "alt" blocks to show different scenarios:
 - **Authorized/Paid:** The PMS records entry in the DB, updates occupancy, and grants access.

- **Unpaid/Full:** The PMS returns a denial message to the driver.
- **Unauthorized:** The PMS denies entry and sends a message to the Security Staff.

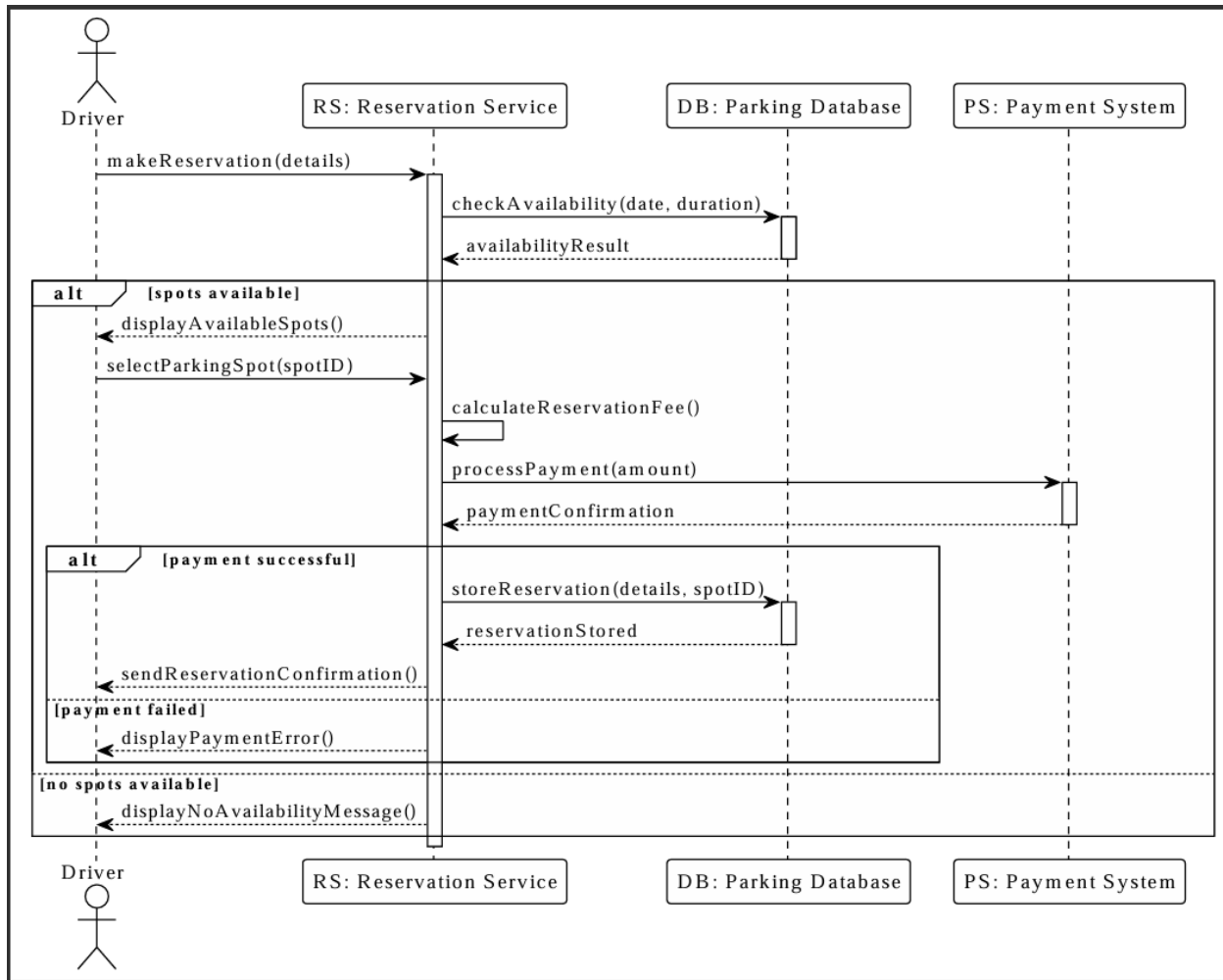
8. Sequence Diagram: Vehicle Exit



This diagram details the object interactions required to process a vehicle leaving the lot.

- **Interaction:** The Driver requests an exit. The PMS retrieves the session from the DB and calculates the fee.
- **Payment Verification:** The PMS contacts the Payment System (PS) to verify if the fee is settled.
- **Alt Frames:**
 - **Payment Completed:** The PMS tells the DB to record the exit time and release the spot, then approves the exit for the Driver.
 - **Payment Not Completed:** The PMS prompts the Driver to pay and simultaneously alerts Security Staff.

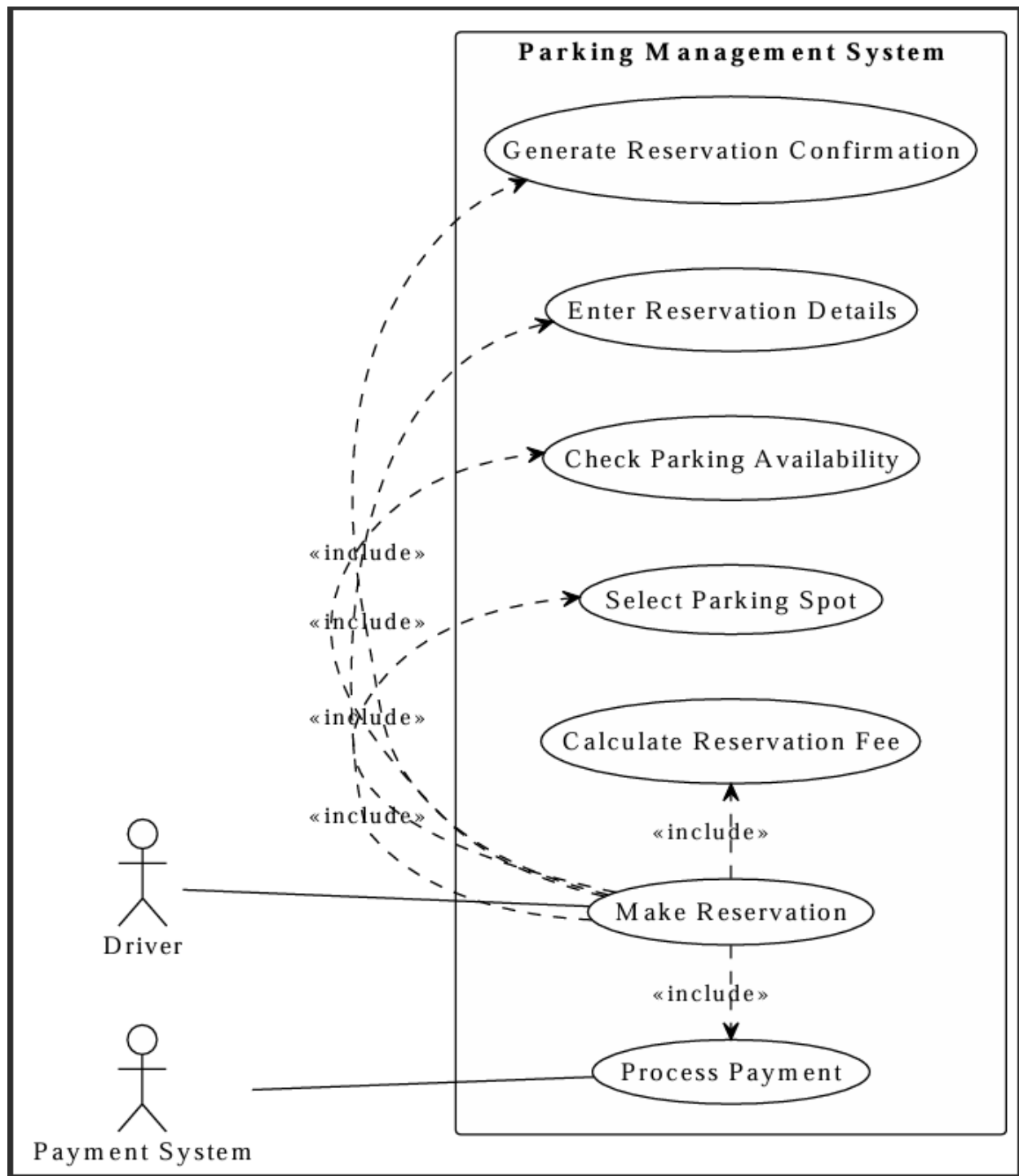
9. Sequence Diagram: Make Reservation



This diagram shows the technical message exchange required to create a new reservation.

- **Interaction:** The Driver sends details to the Reservation Service (RS). The RS queries the DB for availability.
- **Flow:** If spots are available, the RS displays them. The driver selects a spot, and the RS calculates the fee.
- **Payment:** The RS sends a payment request to the Payment System (PS).
- **Alt Frames:**
 - **Success:** If the PS confirms payment, the RS stores the reservation in the DB and sends a confirmation to the Driver.
 - **Failure:** If payment fails or no spots are available, the RS returns an error message to the Driver.

Use Case Description 1: Make Reservation



Item

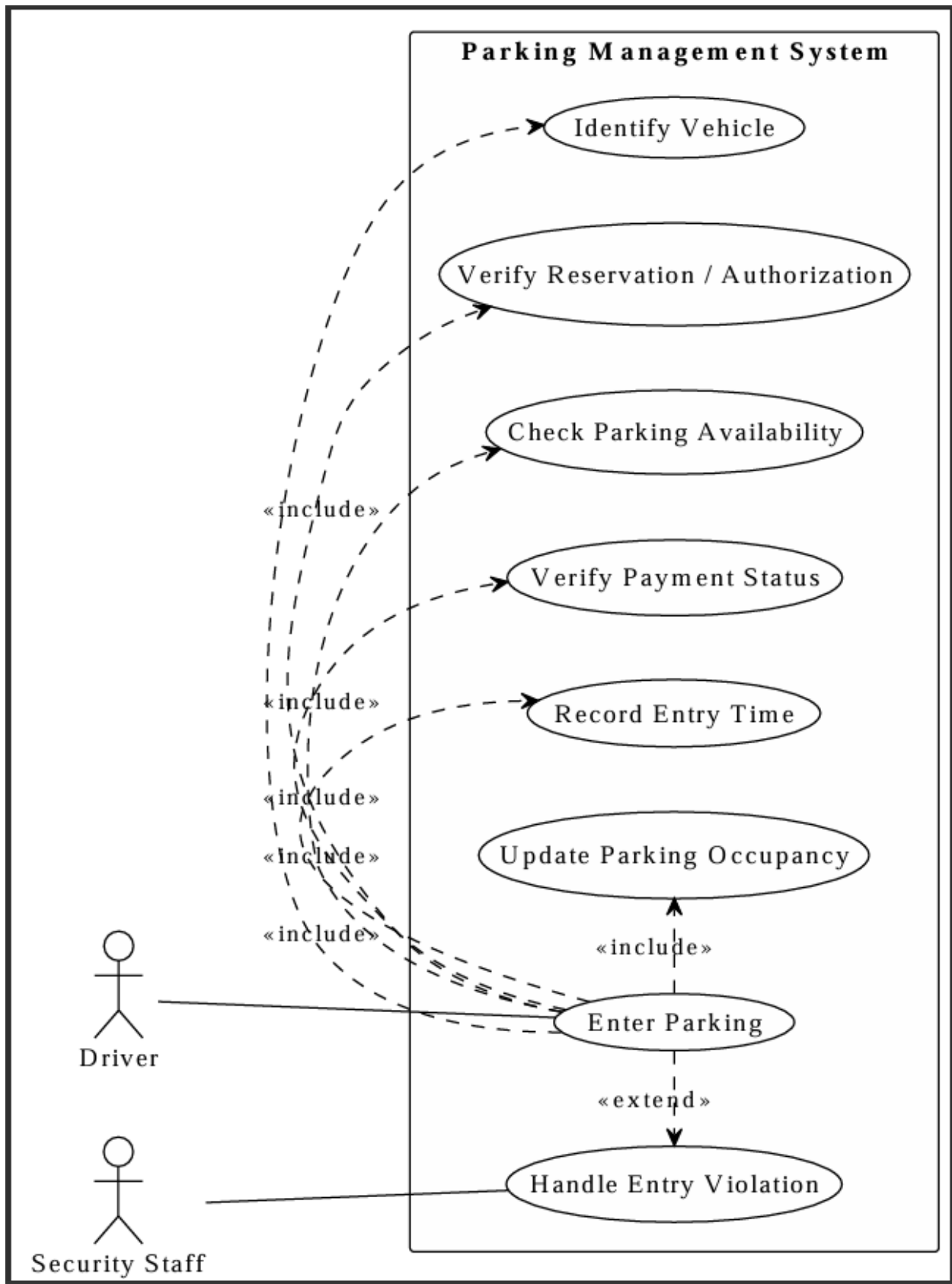
Description

Use Case Name

Make Reservation

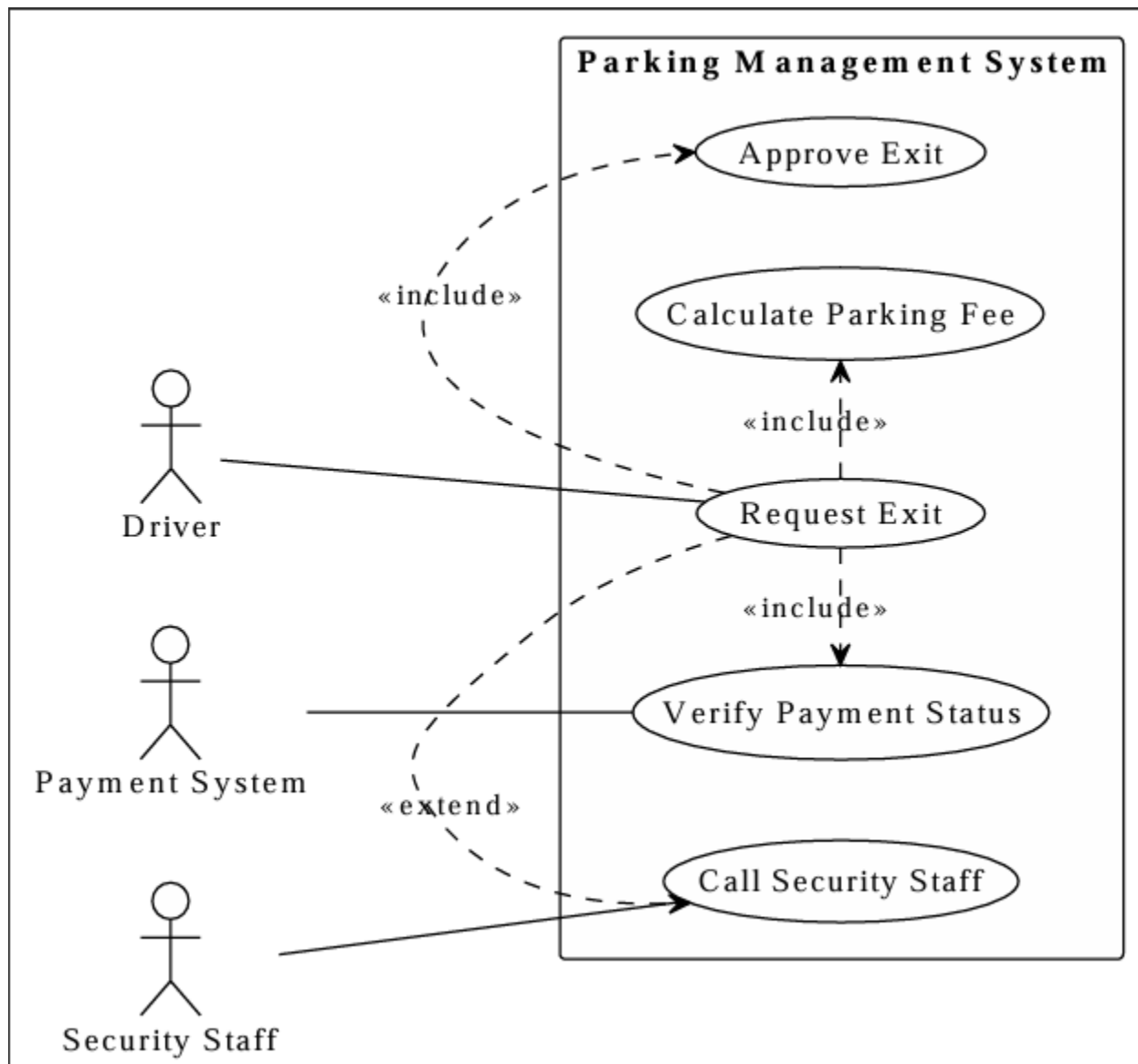
Item	Description
Primary Actor	Driver
Secondary Actor(s)	Payment System
Goal	To allow the driver to reserve a parking spot for a specific date and duration
Preconditions	- Driver is registered and logged in - Parking system is available
Postconditions	- Reservation is created and stored - Parking spot is reserved - Confirmation is sent to the driver
Main Scenario	1. Driver selects "Make Reservation" 2. Driver enters reservation details (date, duration) 3. System checks parking availability 4. Driver selects an available parking spot 5. System calculates the reservation fee 6. System processes payment via Payment System 7. System generates reservation confirmation
Alternative Scenarios	- 3a. No parking spots available: system displays unavailability message - 6a. Payment fails: system displays payment error and reservation is not created

Use Case Description 2: Enter Parking (Vehicle Entry)



Item	Description
Use Case Name	Enter Parking
Primary Actor	Driver
Secondary Actor(s)	Security Staff
Goal	To allow an authorized vehicle to enter the parking facility
Preconditions	- Driver has a valid reservation or authorization - Parking system is operational
Postconditions	- Entry time is recorded - Parking occupancy is updated - Vehicle is granted access
Main Scenario	1. Driver arrives at parking entry 2. System identifies the vehicle 3. System verifies reservation or authorization 4. System checks parking availability 5. System verifies payment status 6. System records entry time 7. System updates parking occupancy 8. System grants entry access
Alternative Scenarios	- 3a. Unauthorized vehicle: entry is denied and call security staff - 5a. Unpaid fees: entry is denied and call security staff - Violation detected: system calls security staff

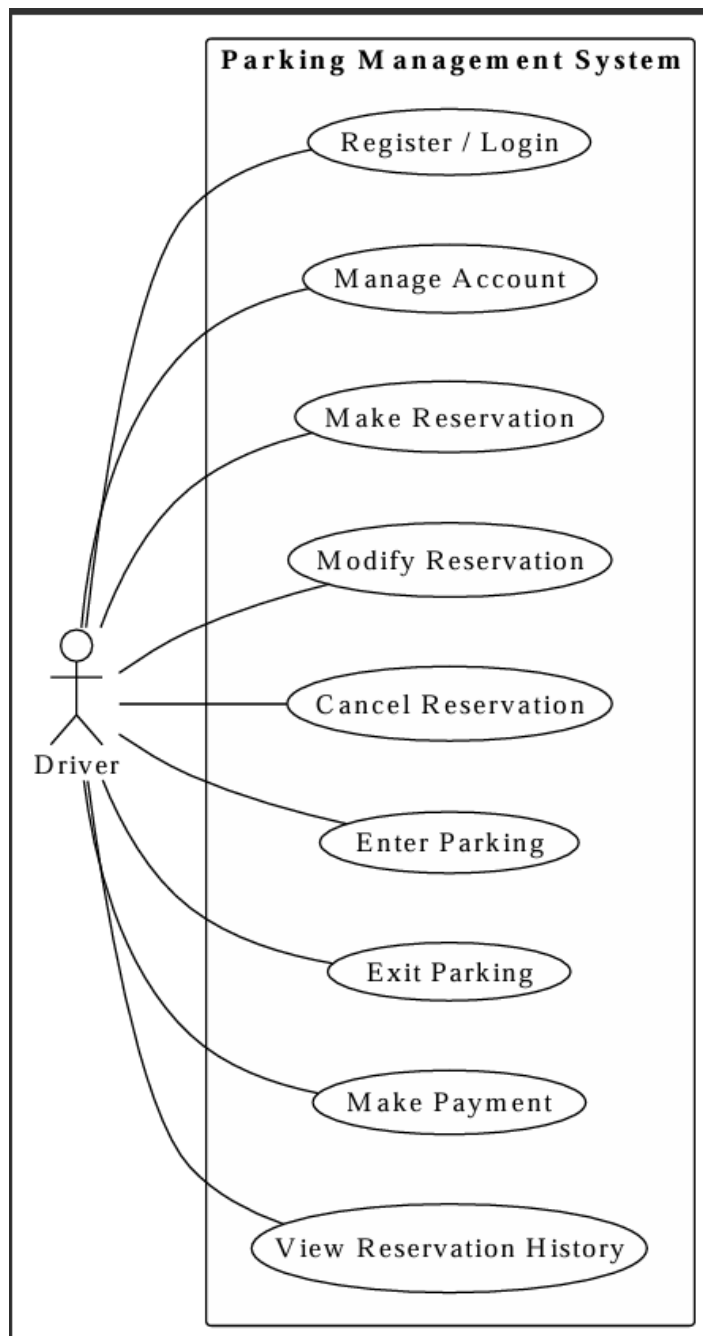
Use Case Description: Exit Parking



Item	Description
Use Case Name	Exit Parking
Primary Actor	Driver
Secondary Actor(s)	Payment System, Security Staff
Goal	To allow a parked vehicle to exit the parking facility after settling any required fees

Item	Description
Preconditions	- Vehicle is currently parked in the facility - Parking system is operational
Postconditions	- Exit time is recorded - Parking fee is settled (if required) - Parking spot is released - Vehicle exits the parking facility
Main Scenario	1. Driver arrives at the parking exit 2. Driver requests exit 3. System calculates parking fee 4. System verifies payment status via the Payment System 5. System approves exit 6. System records exit time 7. System releases the parking spot 8. Vehicle exits the parking facility
Alternative Scenarios	- 4a. Payment not completed: system requests payment and denies exit until payment is completed and call security staff - Payment failure: system denies exit and call security staff - Violation or issue detected: system calls security staff

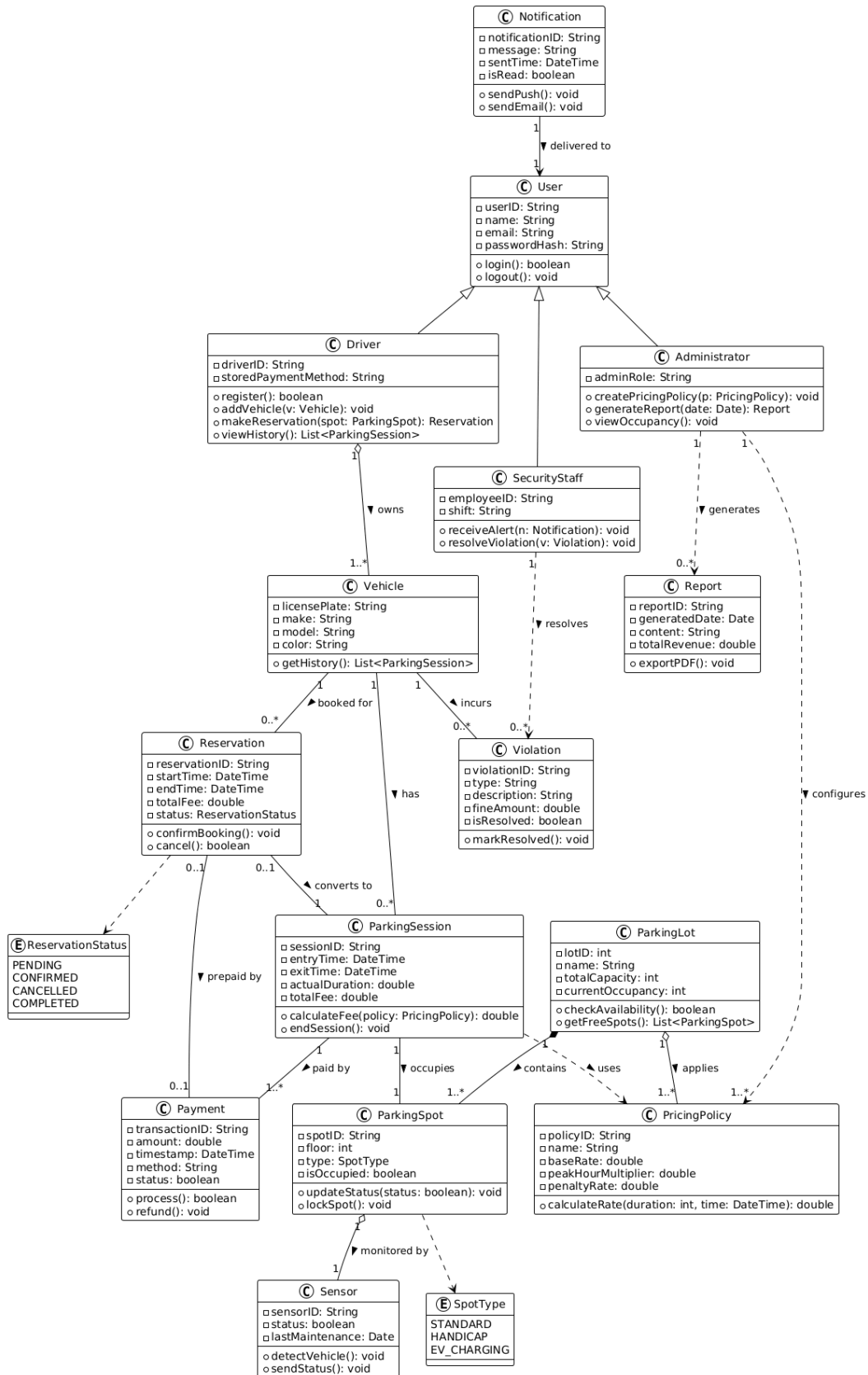
Use Case Description: Driver System Interaction (Composite Use Case)



Item	Description
Use Case Name	Driver System Interaction
Primary Actor	Driver

Item	Description
Secondary Actor(s)	Payment System, Security Staff
Goal	To allow the driver to interact with the parking management system to manage reservations, parking entry and exit, payments, and account information
Preconditions	- Driver has access to the parking management system - System is operational
Postconditions	- Driver actions are completed successfully (reservation, entry, exit, or account updates) - System records are updated accordingly
Main Scenario	1. Driver registers or logs into the system 2. Driver manages account information 3. Driver makes, modifies, or cancels a reservation 4. Driver enters the parking facility 5. Driver makes required payments 6. Driver exits the parking facility 7. Driver views reservation history
Alternative Scenarios	- Payment failure during reservation or exit - Reservation modification or cancellation request - Entry or exit denied due to unpaid fees or violations

Class Diagram: Parking Management System



Class Diagram Explanation

The Class Diagram models the static architecture of the Parking Management System, utilizing object-oriented principles to define the data structure.

1. Inheritance Hierarchy The system uses Generalization for the User class. The Driver, Administrator, and SecurityStaff classes all inherit common attributes (like userID and login credentials) from the User class. This allows them to share core functionality while maintaining their own specific permissions and methods.

2. Structural Relationships The diagram defines two types of ownership. First, there is a Composition relationship between ParkingLot and ParkingSpot. This represents a strong bond where a parking spot cannot exist independently of a parking lot. Second, there are Aggregation relationships. These exist between a Driver and a Vehicle (since a driver owns a vehicle, but they are separate entities) and between a ParkingSpot and a Sensor (representing the hardware attached to the spot).

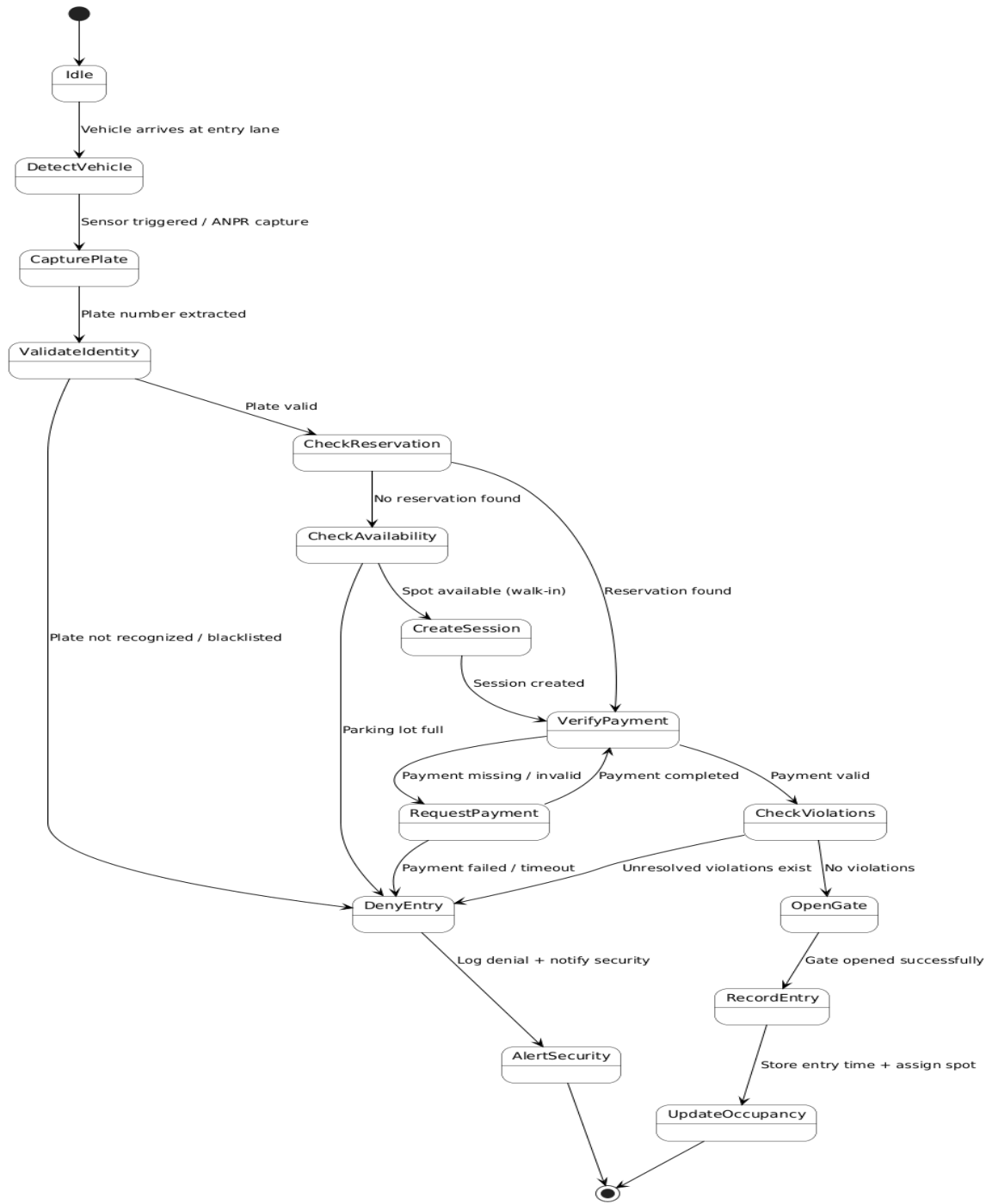
3. Key Subsystems

- **Reservation vs. Session:** The system distinguishes between a future "Reservation" and an active "ParkingSession." A reservation is a booking made in advance, which converts into a session when the driver actually arrives and enters the lot.
- **Financials:** The PricingPolicy class allows the Administrator to configure dynamic rates (like peak hours), which the ParkingSession uses to calculate the final Payment.
- **Violation Management:** The SecurityStaff interacts with Violation objects to track and resolve infractions linked to specific Vehicles, ensuring security protocols are enforced.

State Diagram 1: Vehicle Entry / Entry Gate Controller

This diagram models the **vehicle entry process**, where the system detects an arriving vehicle, captures and validates the license plate, checks reservation or availability, verifies payment and violations, and grants access by opening the gate. If any validation fails, entry is denied and security is notified.

State Diagram: Vehicle Entry (Entry Gate Controller)



#	Current State	Stimulus Event	/ Condition (Guard)	Next State	System Response / Output

1	Idle	Vehicle arrives at entry lane	—	DetectVehicle	Start monitoring entry process
2	DetectVehicle	Sensor triggered (loop detector/camera trigger)	—	CapturePlate	Activate ANPR capture
3	CapturePlate	Image captured + OCR processing	Plate successfully extracted	ValidateIdentity	Extract plate ID and prepare validation request
4	CapturePlate	OCR/ANPR fails	Plate not readable	DenyEntry	Deny entry, log “plate unreadable”
5	Validate Identity	Validation request to DB	Plate is valid/known	Check Reservation	Retrieve driver profile/permission
6	Validate Identity	Validation result returned	Plate blacklisted / unauthorized	DenyEntry	Deny entry, log reason (blacklist/unauthorized)
7	Validate Identity	DB lookup fails	DB/service not available	DenyEntry	Fail safe: deny entry, log system error
8	Check Reservation	Reservation lookup	Reservation exists & within time window	VerifyPayment	Load reservation details and expected fee/status

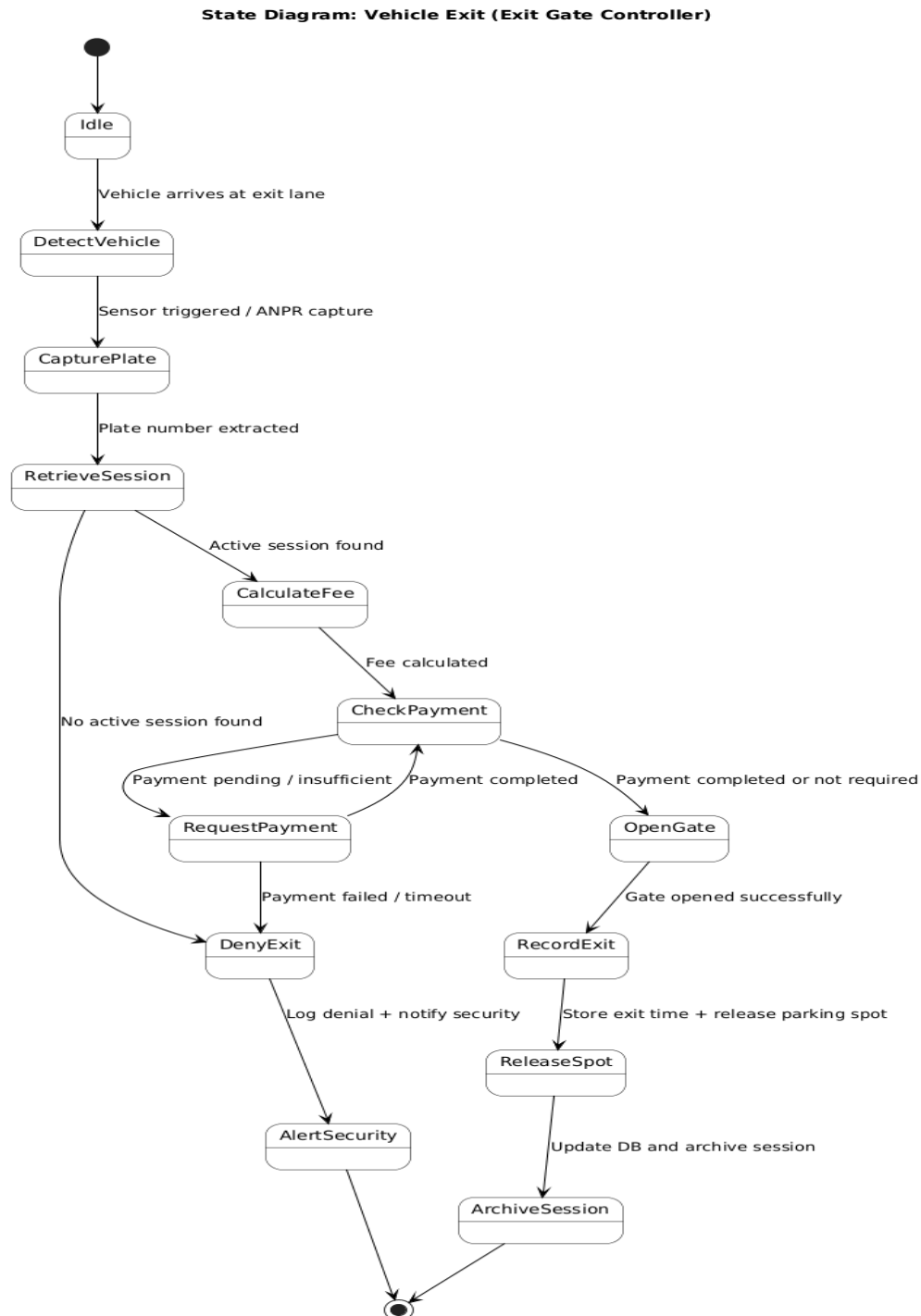
9	Check Reservation	Reservation lookup	No reservation found	Check Availability	Proceed as walk-in request
10	Check Reservation	Reservation lookup	Reservation exists but expired/too early	DenyEntry	Deny entry, show “reservation invalid time”
11	Check Availability	Capacity check	Lot full / no available spots	DenyEntry	Display “Parking Full” + log denial
12	Check Availability	Capacity check	Spot available	CreateSession	Allocate spot logically + prepare walk-in session
13	Create Session	Session creation event	Session created successfully	VerifyPayment	Create active session record (entry pending)
14	Create Session	Session creation fails	DB error / allocation error	DenyEntry	Deny entry, log creation failure
15	VerifyPayment	Payment status check	Already paid / valid payment	CheckViolations	Confirm payment receipt/reference
16	VerifyPayment	Payment status check	Payment missing / invalid / pending	Request Payment	Prompt driver to pay (app/kiosk/card)

17	VerifyPayment	Payment gateway unreachable	Timeout / service down	Request Payment	Offer retry/payment alternative, start timeout timer
18	Request Payment	Payment completed	Success	VerifyPayment	Re-check payment and update transaction status
19	Request Payment	Payment attempt	Failed	Request Payment	Show failure reason, allow retry (limited attempts)
20	Request Payment	Payment timeout	Max wait exceeded	DenyEntry	Deny entry, cancel pending session/reservation hold
21	Request Payment	User cancels payment	Cancel action	DenyEntry	Deny entry and release any reserved capacity/hold
22	Check Violations	Violation check	No unresolved violations	OpenGate	Approve access and send gate-open signal
23	Check Violations	Violation check	Unresolved violation exists	DenyEntry	Deny entry, log violation reason
24	OpenGate	Gate actuator feedback	Gate opened successfully	RecordEntry	Continue entry logging

25	OpenGate	Gate actuator failure	Gate stuck / hardware fault	AlertSecurity	Log fault + notify security/maintenance
26	RecordEntry	Store entry event	Record success	Update Occupancy	Save entry time, link to session/reservation
27	RecordEntry	Store entry event	DB write fails	AlertSecurity	Log critical error + notify security
28	Update Occupancy	Update occupancy/session state	Update success	[Final]	Occupancy increased + session becomes active
29	Update Occupancy	Update occupancy fails	DB/service issue	AlertSecurity	Log inconsistency + notify security
30	DenyEntry	Denial executed	—	AlertSecurity	Send denial reason to security dashboard + store audit log
31	AlertSecurity	Security notified	—	[Final]	End process after alert/logging

State Diagram 2 — Vehicle Exit (Exit Gate Controller)

This diagram models the vehicle exit process, where the system identifies the vehicle, retrieves the active session, calculates fees, verifies payment, and opens the exit gate. If validation or payment fails, exit is denied and security is alerted.



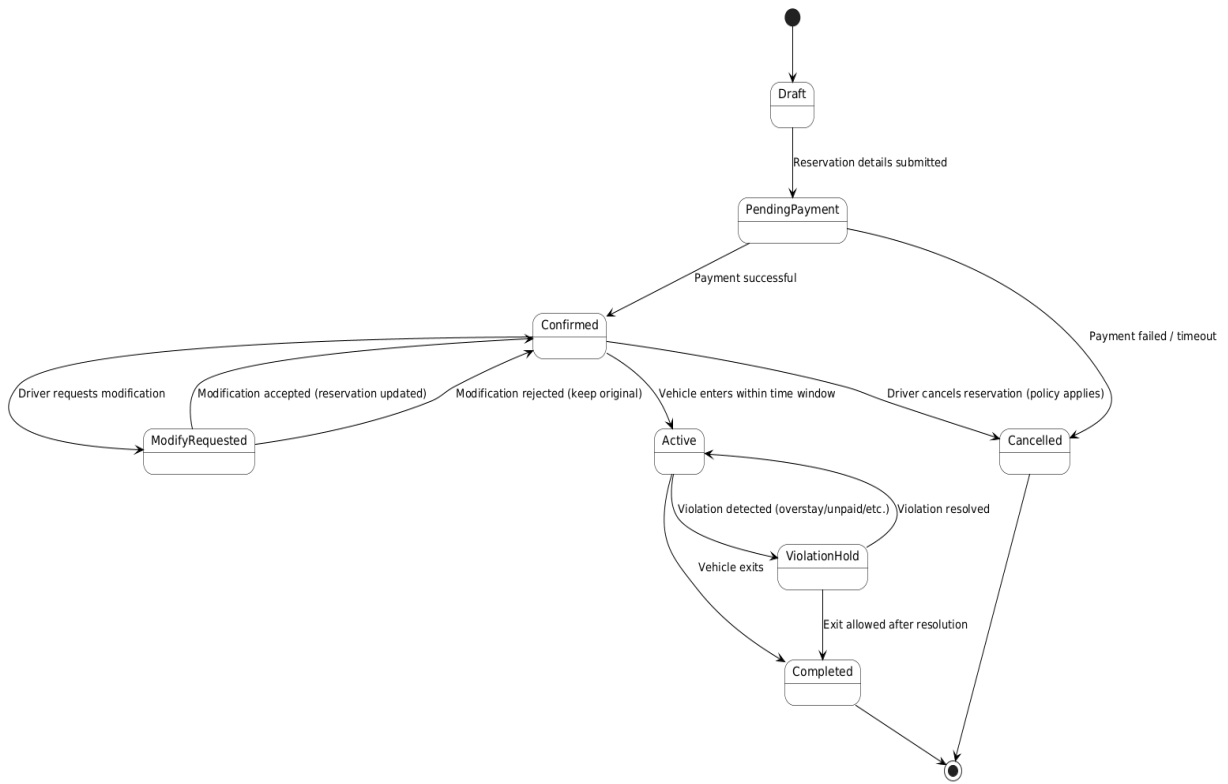
#	Current State	Stimulus / Event	Condition	Next State	System Action
1	Idle	Vehicle arrives at exit	—	DetectVehicle	Start exit workflow
2	DetectVehicle	Sensor triggered	—	CapturePlate	Activate ANPR capture
3	CapturePlate	Plate extraction	Successful	RetrieveSession	Send plate ID to DB
4	CapturePlate	Plate extraction	Failed	DenyExit	Deny exit, log error
5	RetrieveSession	Session lookup	Active session found	CalculateFee	Load session details
6	RetrieveSession	Session lookup	No active session	DenyExit	Deny exit, log violation
7	CalculateFee	Fee computed	—	CheckPayment	Calculate parking charges
8	CheckPayment	Payment verification	Paid / not required	OpenGate	Approve exit
9	CheckPayment	Payment verification	Pending / insufficient	RequestPayment	Request payment

10	RequestPayment	Payment completed	Success	CheckPayment	Re-verify payment
11	RequestPayment	Payment attempt	Failed	RequestPayment	Allow retry (limited)
12	RequestPayment	Payment timeout	Timeout	DenyExit	Deny exit
13	OpenGate	Gate opened	—	RecordExit	Record exit time
14	RecordExit	Exit recorded	—	ReleaseSpot	Release parking spot
15	ReleaseSpot	Spot released	—	ArchiveSession	Archive session & payment
16	ArchiveSession	Archiving done	—	[Final]	End exit process
17	DenyExit	Exit denied	—	AlertSecurity	Notify security

State Diagram 3 – Reservation Lifecycle

This state diagram models the **reservation lifecycle**, starting from creation and payment, through confirmation and optional modification, and activation when the vehicle enters. The reservation ends either by completion after exit or by cancellation, while violations are handled without canceling the original reservation unless necessary.

State Diagram: Reservation Lifecycle (Corrected)



#	Current State	Stimulus / Event	Condition	Next State
1	Draft	Reservation submitted	—	PendingPayment
2	PendingPayment	Payment processed	Success	Confirmed
3	PendingPayment	Payment processed	Failed / timeout	Cancelled
4	Confirmed	Modify request	—	ModifyRequested
5	ModifyRequested	Modification evaluation	Accepted	Confirmed
6	ModifyRequested	Modification evaluation	Rejected	Confirmed
7	Confirmed	Cancellation request	Policy allows	Cancelled
8	Confirmed	Reservation time reached	Vehicle enters	Active
9	Active	Exit event	Vehicle exits	Completed
10	Active	Violation detected	Overstay / unpaid	ViolationHold
11	ViolationHold	Violation resolved	—	Active
12	ViolationHold	Exit after resolution	—	Completed

13	Cancelled	Cancellation finalized	—	[Final]
14	Completed	Completion finalized	—	[Final]